exclurad_py: exact radiative corrections and target straggling for exclusive meson electroproduction

Design, validation and the first π^0 production

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August 24, 2026

Abstract

`exclurad_py` is an event generator for exclusive meson electroproduction $ep \to e' h X$ covering π^0, η, ρ, ϕ and J/ψ in one code, with exact first-order QED radiative corrections built on EXCLURAD and with pre-vertex external radiation in the target. This note is the evidence. The central result is that the generated event density reproduces the exact calculation bin by bin without reweighting: over the azimuth, where the correction swings from 1.15 to 0.92, events and quadrature agree with $\chi^2/N = 0.79$ over eighteen bins. The target radiator is validated against an independent implementation, exclugen, to $0.6\,\sigma$. Four independent samples on three machines, two processor families and two python generations are statistically indistinguishable. For ρ and ϕ the meson mass is drawn from a Breit–Wigner and carried through the kinematics, the t limits and the radiative correction; the resulting mass spectrum agrees with an independently computed prediction at $\chi^2/N = 0.69$. The note states, with equal care, what is *not* yet established: the vector channels still have no external anchor for their radiative correction, and the ρ and ϕ production boxes are run with no inelasticity cut.

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1 Scope and status

The five channels are at different stages, and the difference matters more than any single number in this note.

channel	internal RC	external anchor	B–W mass	status
π^0	validated	published EXCLURAD table	n/a	production running
η	validated	published table via v_{cut}	n/a	next
ρ	vs. own quadrature	none	done, §8	needs $v_{\mathrm{cut}} > 0$
ϕ	vs. own quadrature	none	done, §8	ready to run
J/ψ	vs. own quadrature	none	not needed	box needs care

"Validated against our own quadrature" is a weaker statement than it sounds: generator and reference share the same `rc_factor`, so an error inside that function is invisible to the comparison. It catches sampling errors, which are the common ones, and cannot catch a wrong core. Only π^0 and η have something outside the code to lean on. This is why they come first.

2 The radiative correction

EXCLURAD computes, in the Bardin–Shumeiko formalism, the observed cross section for $ep \to e'h + X_{\mathrm{miss}}$ where the undetected system is reconstructed by missing mass. The variable is the inelasticity

$$v \;=\; M_X^2(ep \to e'h) - m_U^2, \tag{1}$$

zero for a non-radiative event. With an exclusivity cut $v < v_{\mathrm{cut}}$,

$$\sigma_{\mathrm{obs}}(v_{\mathrm{cut}}) \;=\; \sigma_{\mathrm{Born}}\,\big(1 + \delta(v_{\mathrm{cut}})\big), \tag{2}$$

and δ depends on all four hadronic variables and on the cut. Inside the code the same quantity is stored as the multiplicative factor $\eta = \sigma_{\mathrm{obs}}/\sigma_{\mathrm{Born}}$; the note quotes δ in that convention, so $\delta \approx 0.9$ means a 10 % suppression, not a 90 % one.

The soft and hard parts are separated at an arbitrary v_{min}. Nothing physical may depend on it, and §6 shows that nothing does.

3 How the correction is extracted from events

Two runs of the same card differing only in `iborn` give

$$\delta(\mathrm{bin}) = \frac{N_{\mathrm{rad}}(\mathrm{bin})/N_{\mathrm{rad}}}{N_{\mathrm{born}}(\mathrm{bin})/N_{\mathrm{born}}} \times \frac{\sigma_{\mathrm{obs}}}{\sigma_{\mathrm{Born}}}. \tag{3}$$

This is the whole method. Both samples come from the same generator, so the ratio isolates the radiative machinery. Born events cost about a tenth of a radiative one, so the Born sample is made five times larger and contributes about a fifth of the variance.

4 Proof: the correction over the azimuth

The azimuth is the sharpest test available. The correction varies more over ϕ than over any other variable, and ϕ is where a convention error — Trento or not, which hadron defines the plane, the sign of σ_{LT} — shows up. Integrated quantities hide exactly these mistakes.

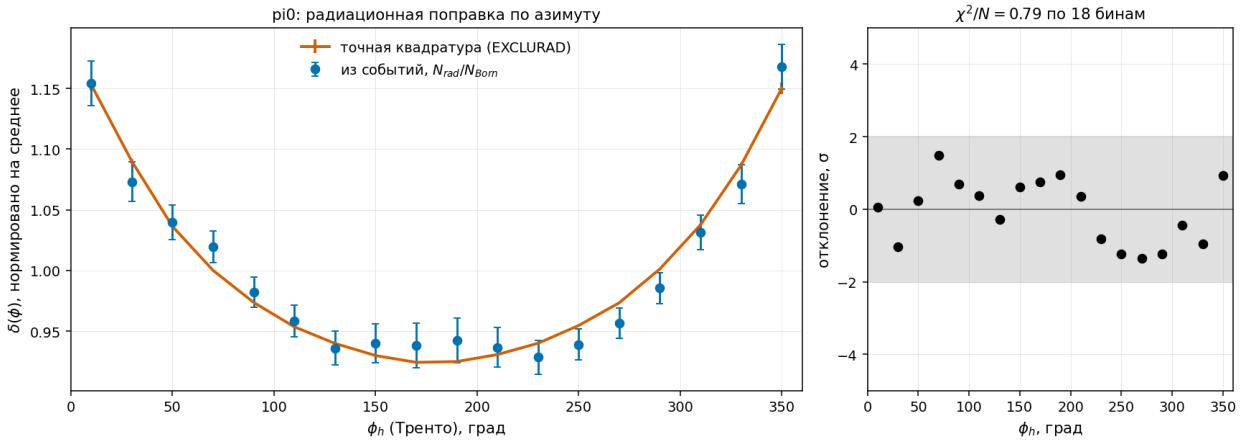


Figure 1: π^0, $E = 10.6$ GeV, Q^2 1.5–2.5, W^2 5.25–6.75, $|t|$ 0.17–2.0, $v_{\text{cut}} = 0.20$ GeV^2, no target radiator. Left: the radiative correction over the azimuth obtained two independent ways — from the events as the ratio of two histograms (points), and from direct integration of the Bardin–Shumeiko formulas at the same kinematic points (line). Right: the difference in standard deviations. 100k radiative and 500k Born events; the quadrature uses 20k points and carries its own error, which is included in the pulls.

Figure 1 is the central result. The two determinations share no machinery: one counts events, the other integrates formulas. They agree with $\chi^2/N = 0.79$ over eighteen bins, the largest deviation being $1.5\,\sigma$, across a correction that swings by a quarter.

The practical consequence is that an analysis binning our events in ϕ recovers the exact correction by construction — no reweighting, no correction table.

ϕ (deg)	from events	from quadrature	σ
10	1.1542 ± 0.0185	1.1532 ± 0.0046	0.1
90	0.9822 ± 0.0125	0.9737 ± 0.0003	0.7
170	0.9379 ± 0.0184	0.9242 ± 0.0005	0.7
250	0.9390 ± 0.0127	0.9546 ± 0.0003	-1.2
350	1.1678 ± 0.0185	1.1501 ± 0.0048	0.9

5 The same in the other variables

Over the full production box the agreement with quadrature holds on every axis. Measured on the 100k union, maximum deviation in standard deviations over twelve bins per axis:

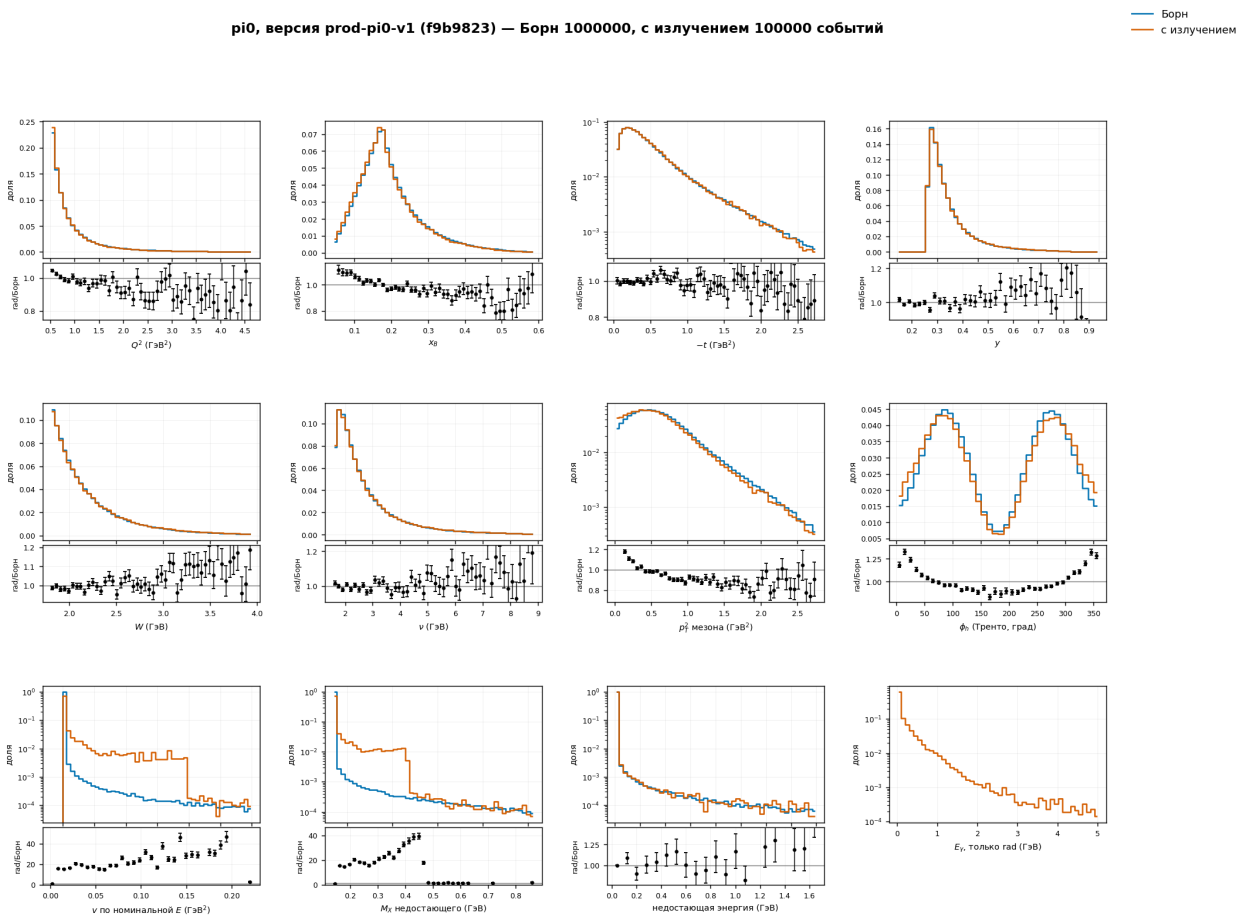


Figure 2: π^0 production sample, radiator on. Upper half of each cell: Born and radiative distributions normalised to unit area; lower half: their ratio. The ratio is the correction as a function of that variable. Note the inelasticity and missing-mass panels: the Born tail stops at the cut while the radiative one extends, which is the radiative tail itself.

channel	Q^2	W	$\lvert t\rvert$	ϕ
π^0	1.8	1.9	1.7	2.0
η	2.0	1.6	1.9	2.6
ρ	2.7	1.5	2.8	1.6
ϕ	2.1	2.7	1.4	1.9
J/ψ	2.9	3.0	1.7	1.9

At twelve bins the worst of twelve is expected around $2.5\,\sigma$ purely by chance, so these numbers are what agreement looks like.

6 Independence of the soft/hard separator

$v_{\min}$ splits the tail integral into an analytic soft part and a numerical hard part. Moving it by two orders of magnitude moves half the contribution from one to the other, and the total must not notice.

$v_{\min}$ (GeV^2)	σ_{obs} (nb)	hard fraction
10^{-5}	0.026400 ± 0.000088	51.3 %
10^{-4}	0.026121 ± 0.000068	42.1 %
10^{-3}	0.026236 ± 0.000087	32.5 %

The hard fraction changes by a factor of 1.6; the cross section is constant to 1.1 %, with the largest pairwise difference at $2.5\,\sigma$. There is no monotonic trend — the middle value lies below both ends — so this looks like statistics rather than a residual dependence. It is honest to say that $2.5\,\sigma$ is "not caught" rather than "clean"; separating a half-percent systematic from a fluctuation needs four times the statistics.

7 External radiation in the target

An electron crosses part of the target before interacting and radiates on the way in. Geant4 cannot supply this: it starts at the vertex and takes the beam energy as given, so the incoming leg is the generator's responsibility and nothing in the CLAS12 chain was providing it. The outgoing leg is deliberately *not* modelled here — Geant4 does it, and doing it twice is worse than not doing it at all.

The sampling follows exclugen and aao_rad: the vertex is uniform over the cell, $bt = (4/3)(v_z - v_{z,\min})/X_0$ with $X_0(\mathrm{LH_2}) = 890.4$ cm, and the surviving energy fraction is $u^{1/bt}$. Against the analytic straggling law over 400k draws: $P(\text{loss} > 1\,\%)$ is 1.674 % against 1.704 %, and $P(\text{loss} > 10\,\%)$ is 0.818 % against 0.857 %.

Validation against an independent implementation. The observable effect is that events are pushed out of the exclusivity window, because an analysis reconstructs the missing mass with the *nominal* beam energy. Running exclugen twice, with and without its own radiator, over six independent pairs of 600k events gives a shift of 0.873 ± 0.112 % in the cut efficiency. Our generator, measured the same way:

source	fraction above the cut	events
mac mini	$0.865 \pm 0.047\,\%$	40k
JLab farm	$0.771 \pm 0.031\,\%$	80k
combined	$\mathbf{0.800 \pm 0.026}\,\%$	
exclugen (reference)	$0.873 \pm 0.112\,\%$	6×600k

Agreement at $0.6\,\sigma$, and with the radiator off, exactly zero events out of forty thousand lie above the cut. This is the first piece of our physics checked against somebody else's code rather than against our own quadrature.

One measurement lesson is worth recording. The scatter between exclugen pairs (0.275 %) exceeds the binomial error of a single pair (0.177 %), because exclugen rebuilds its sampling ceiling from a prescan each run. The first two pairs gave -1.20 and $-0.44\,\%$ and disagreed by three standard deviations; both were correct and both were useless alone.

What this means for the user of the sample. With the radiator on, $v_{\rm cut}$ is applied in *true* variables. About 0.8 % of the delivered events will fail an exclusivity cut made with the nominal beam energy — for Born events too, since the electron loses energy before the vertex whatever happens there. The sample is nevertheless complete:

$$v_{\rm rec} - v_{\rm true} = 2\,\Delta E\,(E_X - p_{X,z}) \;\geq\; 0, \tag{4}$$

so external loss can only move events out of the window, never into it.

8 The Breit–Wigner mass of the vector mesons

Until this pass every ρ was produced at exactly 775.3 MeV. Its width is 150 MeV — nineteen percent — so a fixed mass puts every $\pi^+\pi^-$ pair in one bin where the experiment sees a broad resonance. For ϕ, $\Gamma = 4.27$ MeV and the peak itself is near the resolution; the mechanism is nonetheless the same, and the ϕ serves as the control.

The mass does not stay in the lineshape. It sets the W threshold, it moves the t limits, it enters every four-vector, and through the kinematics it enters the region where the radiative correction is evaluated. It is therefore drawn per trial, at the same place in the loop where the beam energy is drawn, and carried through: the event kinematics, $v_{\rm max}$, the t limits, $\sigma_{\rm Born}$, `rc_factor` and the decay.

The sampler. A transcription of `sample_bw` from `diffrad_vm.f90`, deliberately without improvement: a reference exists, and a difference from it would be a transcription error rather than a better answer. The substitution $M^2 = M_0^2 + M_0\Gamma\tan\theta$ with uniform θ gives an exact constant-width Breit–Wigner with no rejection; for ρ a rejection on the running p-wave width $\Gamma(M) = \Gamma_0\,(p/p_0)^3\,(M_0/M)$ is applied on top, with the envelope normalised to its scanned maximum because the constant-width form is not everywhere above the running one. Against the Fortran over 200k draws each: $\chi^2/N = 0.80$ for ρ and 0.73 for ϕ.

Nothing else moved. With the draw disabled the generator reproduces its previous output byte for byte — the same 2442 lines, only the provenance header differing. This is the first thing to establish when a quantity that was a constant becomes a per-event variable, and it is cheap: the draw consumes no random numbers when it is off, so the stream is untouched.

The mass spectrum is predicted, not merely plausible. In the sample the mass is *not* distributed as a Breit–Wigner. The drawn mass then takes part in the acceptance, so the density is

$$p(M) \;\propto\; \frac{\mathrm{BW}(M)}{N(W^2)} \int \mathrm{d}V \; \sigma_{\mathrm{Born}}(V; M), \tag{5}$$

and the mean ρ mass moves from 0.894 GeV for the bare sampler to 0.847 in the events. A shift of that size is explained equally well by a correct acceptance and by an error in the threading, so it has to be predicted. The right-hand side of (5) is evaluated by uniform quadrature over the box, with no majorant, no accept–reject and no generator loop — what is being tested is the sampling. The Born cross section is the same function the generator calls; its physics is checked elsewhere and is common to both sides here.

The factor $N(W^2)$ is not a detail. The ceiling of the draw is $\min(\sqrt{W^2} - m_p,\ M_0 + 5\Gamma)$ and depends on the event, so in the low-W^2 corner the resonance is truncated and the truncated Breit–Wigner is normalised on its own interval. Events from that corner therefore enter the prediction with a larger weight.

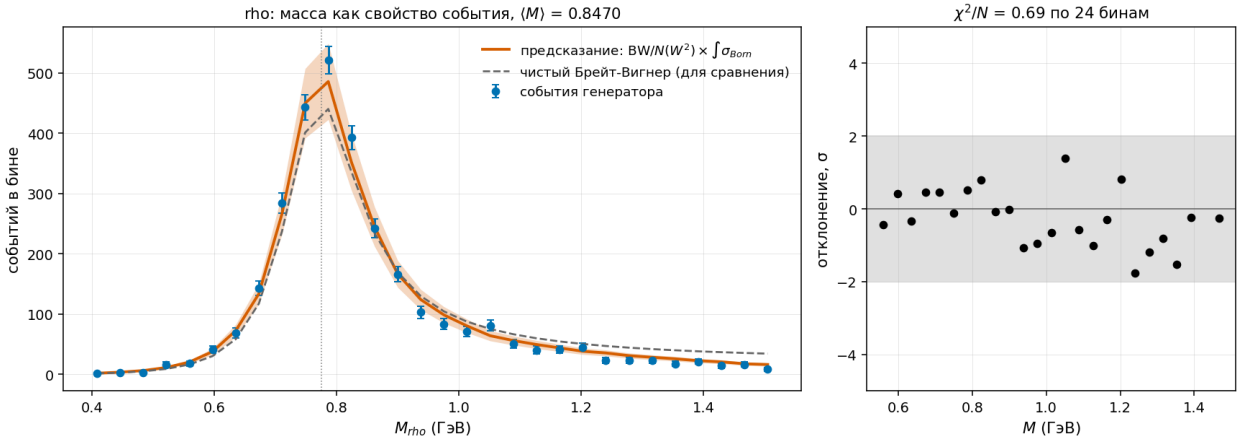


Figure 3: ρ, Born, 10.6 GeV, 3000 events. Left: the mass spectrum in the events (points) against the prediction (5) with its quadrature error (band), and a pure Breit–Wigner for comparison (dashed). The events sit visibly below the pure lineshape above the pole, because the cross section falls with mass, and the prediction follows them down. Right: the deviation in standard deviations. $\chi^2/N = 0.69$ over 24 bins; for ϕ, 0.52 over 18, where the two curves are indistinguishable, which is the right answer for a 4 MeV resonance.

Two things the threading broke, and how they were found. Both were found by the closure test rather than by inspection, which is the argument for having written it.

The first is bookkeeping. The total cross section is estimated as $(n_{\mathrm{acc}}/n_{\mathrm{trials}})$ times the majorant integral, and 37 % of ρ trials now die on t limits that move with the mass. The first version discarded those before incrementing the trial counter, which would have inflated the ρ cross section by half. A proposal drawn from the majorant has been spent and must appear in the denominator with zero weight, exactly as a kinematic cut does. With the counting corrected, Breit–Wigner lowers the ρ cross section by 8 %: 0.9213 from the generator against 0.9346 ± 0.0185 from direct quadrature over the same box.

The second is physics. `rc_factor` took the nominal mass. Across the ρ range the correction

varies by 7–25 % depending on the point, so an event generated at 1.35 GeV was being corrected as though it were 0.775. The function now takes the event's mass.

M_ρ (GeV)	0.45	0.60	0.775	0.90	1.10	1.35
δ at $Q^2 = 2$, $W^2 = 8$, $t = -0.5$, $\phi = 3$	1.143	1.119	1.084	1.056	1.003	0.914

What it changes in the observables. The peak has to appear; what matters is what the mass drags with it, and in particular the daughter tracks, since those and not the meson are what a detector sees. For ρ: $|t|$ shifts by 5.8 %, the π^+ momentum by 1.5 %, its polar angle by 4.2 % and the pair opening angle by 7.1 %. For ϕ everything stays within half a percent — the correct answer for a 4 MeV resonance, and worth showing beside ρ because it demonstrates that the draw carries nothing spurious with it.

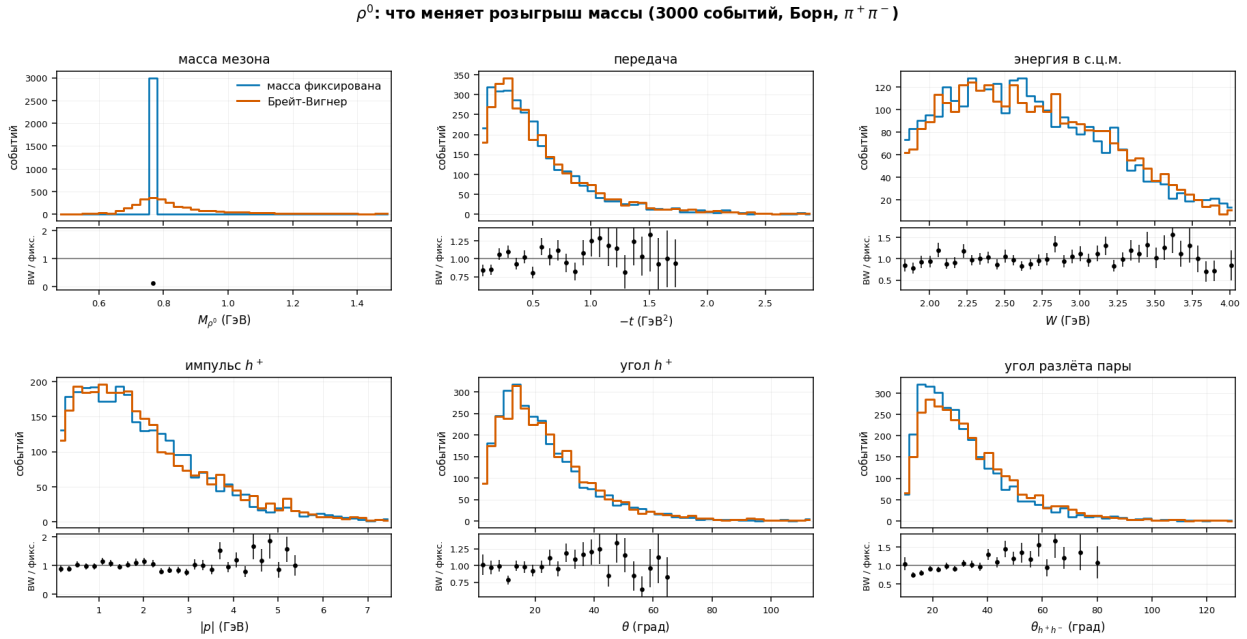


Figure 4: ρ: mass fixed at the pole (blue) against mass drawn from the Breit–Wigner (orange). Same seed, same card; the only difference is the mass. Lower panels: the ratio of the two. The missing-mass panel is omitted because at Born it is the proton mass in every event.

Two errors in the check itself are worth recording, because both were invisible at 400k draws and obvious at 1.6M, which is the argument for testing convergence in N rather than at one sample size. The factor M/M_0 in the analytic density belongs only to the running-width branch, where it arrives with the rejection; applied to ϕ as well it biases the mean mass by 2 %. And a 4 MeV resonance binned at 0.9 MeV must be integrated over the bin: taking the density at the bin centre gives $\chi^2/N \approx 2.4$ and looks exactly like a broken sampler.

9 Reproducibility

The worst χ^2/N over ten variables is 1.45 (p_T^2) and the median is about 1.1. Nothing in the chain — processor, library version, interactive versus batch — leaves a trace in the sample. Two further

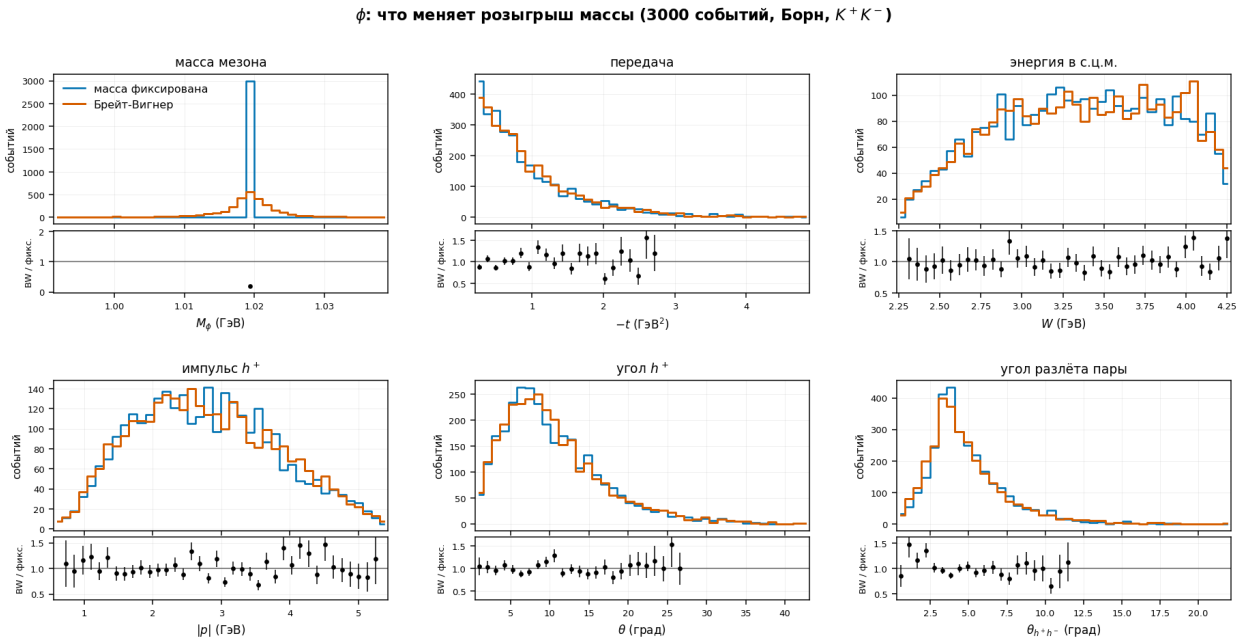


Figure 5: ϕ, the same comparison. Everything within half a percent, as a 4 MeV width requires. A control of this kind is worth as much as the ρ figure: it bounds what the machinery does when it should do nothing.

checks: a run repeated with the same seed reproduces its output byte for byte, and 100k events pooled from two machines contain exactly 100k distinct events.

10 Output integrity

Conservation laws are checked per event rather than inferred from the shape of a distribution: the header's particle count, four-momentum against beam plus target over the status-1 particles, mass shells, the written mass against PDG, decay closure, charge, and the beam-polarisation column.

Four-momentum needs care once the target radiator is on, and the earlier version of this paragraph got it wrong. The event is built at the event's own *reduced* beam energy and the pre-vertex photon is never written — it is absorbed into that reduction — so summing the status-1 particles against the *nominal* beam leaves a residual equal to the loss. Tested that way, every production sample failed: 5.7 % of π^0 events, worst residual 8.9 GeV. The number quoted here before, 1.25×10^{-8} GeV, came from a run with the radiator off and did not describe the delivered samples.

The check now asks the sharper question, and the samples pass it. Where a residual exists it must be a lightlike vector *along the beam* with *positive* energy — what a pre-vertex photon can be and nothing else can. Measured over 600k production events: $|p_x|, |p_y| \leq 9 \times 10^{-11}$, $|E - p_z| \leq 8 \times 10^{-8}$, not one negative energy. A lost particle, a sign error or an unclosed decay would leave a residual that is not collinear, not lightlike, or negative. Elsewhere four-momentum closes to 10^{-6} and the decay to 10^{-10}.

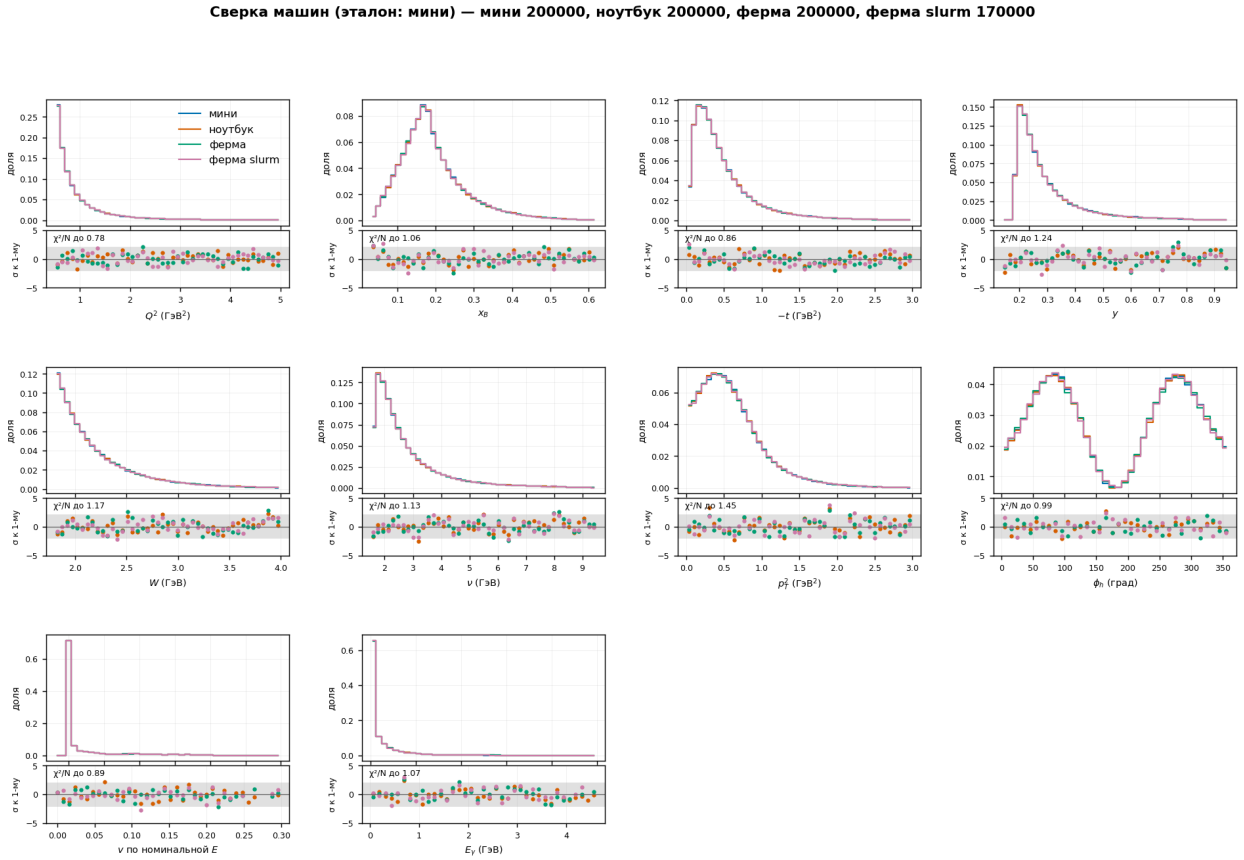


Figure 6: Four independent samples: mac mini (M4), MacBook Pro (M3 Max), the JLab farm interactively and the same farm through slurm. Different seeds, different processors, python 3.9 with numpy 1.23 against python 3.14 with numpy 2.4. Lower panels: deviation from the first sample in standard deviations, grey band $\pm 2\sigma$.

11 What is not established

The vector channels have no external anchor. ρ, ϕ and J/ψ are checked against our own quadrature only. DIFFRAD, the Akushevich code for vector mesons, is on disk and is the natural reference; this comparison has not been made.

The Breit–Wigner has no end-to-end external anchor. The sampler matches `diffrad_vm.f90` and the threading is closed against our own quadrature (§8), but no vector-meson mass spectrum from a *different* code has been compared with ours after the full generation chain. DIFFRAD would supply exactly that, and it is the same missing comparison as the one below.

The η table is built at the nominal mass. The interpolated η serves only as the sampling proposal, and the second stage recomputes `rc_factor` exactly at the event's own mass, so this costs efficiency and not correctness. It was checked rather than assumed: with the draw on, ρ and ϕ show no majorant violations beyond the pre-existing ρ ones, and the largest overweight is smaller with the draw than without it.

ρ and ϕ run with no inelasticity cut. $v_{\mathrm{cut}} = 0$ by decision, meaning the tail runs to the kinematic limit where the diffractive model is extrapolated far outside its fit. With $v_{\mathrm{cut}} = 0$ the ρ correction carries an 8 % slope in Q^2 that the quadrature does not have ($\chi^2/N = 4.04$); at $v_{\mathrm{cut}} = 0.20$ the slope is gone (2.04).

π^0 table interpolation. The interpolation rms is 0.029 against a tolerance of 0.02, and refining does not help: $1000 \to 6144$ nodes moves it from 0.029 to 0.027, while the other four channels sit at 0.002–0.006. Correctness is unaffected — the second stage is exact — but efficiency is.

The exponentiation systematic is measured but not resolved: δ with and without the YFS exponentiation differs by $-0.66\,\%$ (π^0), $-0.48\,\%$ (η), $-1.47\,\%$ (ρ), $-1.44\,\%$ (ϕ) and $-0.09\,\%$ (J/ψ). It should be quoted as a theory systematic.

12 Summary

For π^0 the generator is ready: the event density reproduces the exact calculation over the azimuth to $\chi^2/N = 0.79$, the target radiator agrees with an independent implementation to $0.6\,\sigma$, four samples on three machines are indistinguishable, and the output satisfies the conservation laws event by event. η shares the machinery and needs its own production and its own published-table comparison.

For the vector channels the Breit–Wigner mass is now a property of the event and closes against an independent prediction at $\chi^2/N = 0.69$; ϕ is ready to run, while ρ should first be given a non-zero v_{cut}, since with $v_{\mathrm{cut}} = 0$ its tail reaches kinematics where the diffractive model is extrapolated far outside its fit. What all three still lack is an external anchor for the radiative correction itself, for which DIFFRAD is available and unused. Two of the three findings in §8 — the trial counting and the mass in `rc_factor` — were produced by the closure test rather than by reading the code, which is the case for writing the test before trusting the change.